Indian Institute of Technology Hyderabad
Signal Processing, Fall 2026
Quiz 4, 08.10.2026, 10 points

1. The impulse response of an LTI system with finite impulse response is antisymmetric, i.e., $h[n] = -h[M-n]$ with M even. The impulse response is zero outside the range $0 \leq n \leq M$.
   (a) Derive $H(e^{j\omega})$. Use the antisymmetry property to express $H(e^{j\omega})$ compactly. (2)
   (b) Plot the general form of the phase response of this system. (1)

2. Apply the series expansion method to find the inverse z-transform of $X(z) = \frac{1}{1-az^{-1}}$ with ROC $|z| > |a|$. Explain the steps in your work. (1)

3. Let's work with $h_{lp}[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$ and $h_{hp}[n] = \frac{1}{2}(\delta[n] - \delta[n-1])$ and the z-transform. Specifically, let $h[n] = h_{lp}[n] * h_{hp}[n]$, be a cascade of the low-pass and high-pass filters.
   (a) Find $h[n]$ by inverting $H(z) = H_{lp}(z)H_{hp}(z)$. Justify your answer. (2)
   (b) Traverse the unit circle to predict the DTFT magnitude response of the cascaded system $h[n]$. (2)
   (c) Verify your answer in the previous question by explicitly computing $|H(e^{j\omega})|$. (2)